

Student Name: .....

**2005**  
**TRIAL HIGHER SCHOOL CERTIFICATE**

**MATHEMATICS**  
**Extension 1**



**General Instructions**

Reading Time: 5 minutes  
Working Time: 2 hours

- Attempt all questions
- Start each question on a new page
- Each question is of equal value
- Show all necessary working.
- Marks may be deducted for careless work or incomplete solutions
- Standard integrals are printed on the last page
- Board-approved calculators may be used

### Question One

Marks

a) Find the remainder when  $P(x) = x^3 + 2x - 5$  is divided by  $x - 1$ .

1

b) Use the table of standard integrals to find  $\int \frac{1}{\sqrt{x^2 + 4}} dx$

1

c) Solve the inequality  $\frac{2}{x-5} > 1$

2

d) For the points  $A(-3, 8)$  and  $B(5, -6)$ , find the coordinates of the point  $P$  which divides the interval  $AB$  internally in the ratio 1:3.

2

e) Use the substitution  $u = x - 1$  to evaluate  $\int_2^4 \frac{x}{(x-1)^2} dx$ .

3

f) Solve the equation  $\sin 2\theta = \cos \theta$  for  $0 \leq \theta \leq 2\pi$ .

3

### Question Two ( Start a new page )

Marks

a) Find  $\frac{d}{dx}(x^3 \tan^{-1} 2x)$

2

b) The function  $f(x) = \ln x + 5x$  has a zero near  $x = 0.2$

2

Use one application of Newton's Method to find a better approximation, giving your answer correct to 2 decimal places.

c) For the function  $f(x) = \cos^{-1}(3x)$

i) Find  $f(-\frac{1}{6})$ , expressing the answer in radians in exact form.

1

ii) State the domain and range of this function.

1

iii) Neatly sketch  $y = f(x)$ .

2

d) i) Show that  $\frac{x+1}{x+3} = 1 - \frac{2}{x+3}$

1

ii) You are now given that  $y = \frac{x+1}{x+3}$ , find the equation of the vertical asymptote.

1

iii) Without the use of calculus neatly sketch the graph of  $y = \frac{x+1}{x+3}$  showing all the main features.

2

### Question Three ( Start a new page )

Marks

a) Given that  $\int_0^2 f(t) dt = 5$ , evaluate  $\int_0^1 f(t) dt + \int_1^2 f(t) - 1 dt$ .

2

b) If the line  $y = mx + b$  is 2 units from the origin, prove that  $m^2 + 1 = \frac{b^2}{4}$ .

2

c) Prove, by Mathematical Induction, that for all integers  $n \geq 1$

4

$$1 + 6 + 15 + \dots + n(2n-1) = \frac{1}{6}n(4n-1)(n+1).$$

d) The population  $N$ , of Keystown first reached 25 000 on 1 January 2000. The population of Keystown is set to increase according to the equation

$$\frac{dN}{dt} = k(N - 8000)$$

where  $t$  represents time in years after the population first reached 25 000. On 1 January 2005, the population of Keystown was 29 250.

i) Verify that  $N = 8000 + Ae^{kt}$  is a solution to the above equation where  $A$  is a constant.

1

ii) Find the value of  $A$  and the value of  $k$  correct to 4 decimal places.

2

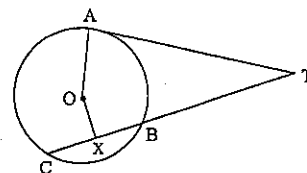
iii) In which year will the population of Keystown reach 50 000?

1

### Question Four ( Start a new page )

Marks

a)



A, B and C are 3 points on a circle centre O. The tangent at A meets CB produced at T. X is the midpoint of BC.

Neatly copy the diagram onto your answer sheet.

i) Without adding any constructions to the diagram prove that AOXT is a cyclic quadrilateral.

3

ii) Hence state why  $\angle AOT = \angle AXT$

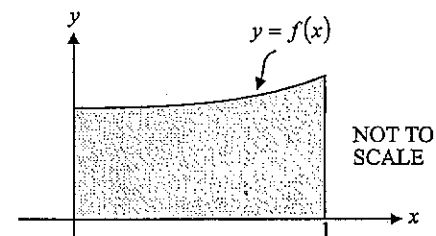
1

b) i) Find the natural domain of the function  $f(x) = \frac{1}{\sqrt{4-x^2}}$ .

1

ii) The sketch below shows part of the graph of  $y = f(x)$ . The area under the curve for  $0 \leq x \leq 1$  is shaded. Find the area of the shaded region.

2



c) If  $\alpha, \beta, \gamma$  are the roots of  $2x^3 + x^2 - x - 2 = 0$ , find the value of:

i)  $\alpha + \beta + \gamma$

1

ii)  $\alpha\beta\gamma$

1

iii)  $\alpha\beta + \alpha\gamma + \beta\gamma$

1

iv) Hence or otherwise, find the value of  $(\alpha-1)(\beta-1)(\gamma-1)$ .

2

### Question Five ( Start a new page )

Marks

- a) Air is being pumped into a spherical balloon at the rate of  $450 \text{ cm}^3/\text{s}$ .

3

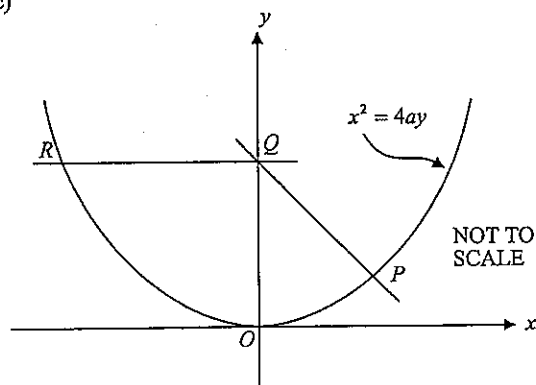
Calculate in exact form the rate at which the radius of the balloon is increasing at the instant when the radius reaches 15cm.  $\left[ V = \frac{4}{3}\pi r^3 \right]$

- b) The area between the curve  $y = \sin x$ , the  $x$ -axis, the lines  $x = 0$  and  $x = \frac{3\pi}{4}$  is rotated about the  $x$ -axis. Find the volume of the solid formed. Express

3

your answer in exact form.

c)



The diagram above shows the graph of the parabola  $x^2 = 4ay$ .

The normal to the parabola at the variable point  $P(2at, at^2)$ ,  $t > 0$ , cuts the  $y$ -axis at  $Q$ . The point  $R$  lies on the parabola in the 2<sup>nd</sup> quadrant.

- Show that the equation of the normal to the parabola at  $P$  is  $x + ty = at^3 + 2at$ .
- Find the coordinates of  $R$  given that  $QR$  is parallel to the  $x$ -axis.
- Let  $M$  be the midpoint of  $RQ$ . Find the Cartesian equation of the locus of  $M$ .

2

2

2

### Question Six( Start a new page )

Marks

- a) A particle moves in a straight line with an acceleration given by

$$\frac{d^2x}{dt^2} = 9(x-2)$$

where  $x$  is the displacement in metres from an origin  $O$  after  $t$  seconds.

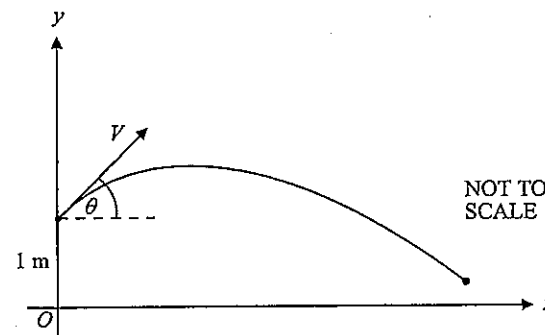
Initially, the particle is 4 metres to the right of  $O$  with a velocity  $v = -6$ .

- Show that  $v^2 = 9(x-2)^2$ .
- Find an expression for  $v$  and hence find  $x$  as a function of  $t$ .

3

3

- b) A boy throws a ball and projects it with a speed of  $V \text{ m s}^{-1}$  from a point 1 metre above the ground. The ball lands on top of a flowerpot in a neighbour's yard.



The angle of projection is  $\theta$  as indicated in the diagram. The equations of motion of the ball are  $\ddot{x} = 0$  and  $\ddot{y} = -10$ .

It has been found that  $y = Vt \sin \theta - 5t^2 + 1$ .

- Show that  $x = Vt \cos \theta$ .
- When the ball is at its maximum height above the ground, it passes directly above a 1.5 metre high fence and clears the fence by 0.5 metres.

1

3

Show that  $V = \frac{\sqrt{20}}{\sin \theta}$

- Find the value of  $V$  given that  $\theta = \tan^{-1} \frac{9}{40}$ .  
Give your answer in  $\text{m/s}$ , correct to 2 decimal places.

2

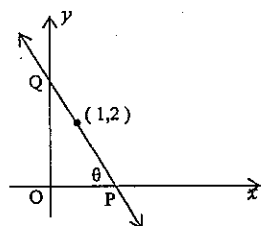
Question Seven( Start a new page )

Marks

a) Find  $\lim_{x \rightarrow 0} \frac{1 - \cos 2x}{3x^2}$

2

b)



A line passes through the point  $(1, 2)$  and meets the  $x$  and  $y$  axes at  $P$  and  $Q$  respectively as shown in the above diagram.  $\angle OPQ = \theta$ .

i) Show that the equation of the line  $PQ$  is given by  $y = \tan \theta - x \tan \theta + 2$ .

2

ii) Show that the area ( $A$ ) of  $\triangle OPQ$  is given by

$$A = \frac{\tan \theta}{2} + 2 + \frac{2}{\tan \theta}.$$

3

iii) Prove that the area is a minimum when  $\tan \theta = 2$ .

4

iv) Hence, find the minimum area.

1

**END OF EXAM**

## STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left( x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left( x + \sqrt{x^2 + a^2} \right)$$

**NOTE:**  $\ln x = \log_e x, \quad x > 0$

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**Question 1.**

a)  $P(1) = -2$

b)  $\ln(x + \sqrt{x^2 + 4}) + C$

c)  $2(x-5) > (x-5)^2$

$$2(x-5) - (x-5)^2 > 0$$

$$(x-5)(7-x) > 0 \Rightarrow 5 < x < 7$$

d)  $x = \frac{3(-3)+1(5)}{1+3}, y = \frac{3(8)+1(-6)}{1+3}$

$\therefore P$  has coordinates  $\left(-1, \frac{9}{2}\right)$ .

e) Let  $u = x-1 \Rightarrow x = u+1$  and  $du = dx$ .  
Also when  $x=4, u=3$  and when  $x=2, u=1$ .

$$\therefore I = \int_1^3 \frac{u+1}{u^2} du \Rightarrow I = \int_1^3 \frac{1}{u} + u^{-2} du$$

$$\therefore I = \left[ \ln u - \frac{1}{u} \right]_1^3 = \left( \ln 3 - \frac{1}{3} \right) - (\ln 1 - 1)$$

$$= \frac{2}{3} + \ln 3.$$

f)  $2 \sin \theta \cos \theta - \cos \theta = 0$

$$\cos \theta (2 \sin \theta - 1) = 0$$

$$\therefore \cos \theta = 0 \text{ or } \sin \theta = \frac{1}{2}$$

$$\therefore \theta = \frac{\pi}{2}, \frac{3\pi}{2} \text{ and } \theta = \frac{\pi}{6}, \frac{5\pi}{6}$$

**Question 2.**

a)

$$\frac{d}{dx}(x^3 \tan^{-1} x) = \tan^{-1} 2x \times 3x^2 + x^3 \times \frac{1}{1+(2x)^2} \times 2$$

$$= 3x^2 \tan^{-1} 2x + \frac{2x^3}{1+4x^2}$$

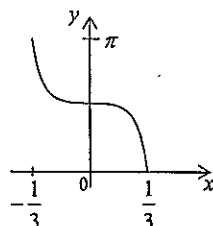
b)  $f(x) = \ln x + 5x \Rightarrow f'(x) = \frac{1}{x} + 5$

$$\therefore x = 0.2 - \frac{\ln(0.2)+1}{10} \Rightarrow x = 0.26$$

c) i)  $\cos^{-1}\left(-\frac{1}{2}\right) = \frac{2\pi}{3}$

ii) D:  $-\frac{1}{3} \leq x \leq \frac{1}{3}$ , R:  $0 \leq y \leq \pi$

iii)

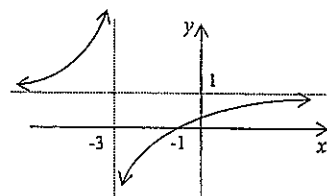


d)  $LHS = \frac{x+3-2}{x+3} = \frac{x+3}{x+3} - \frac{2}{x+3}$

$$= 1 - \frac{2}{x+3} = RHS.$$

ii)  $x = -3$ .

iii)



**Question 3.**

a)  $I = \int_0^1 f(t) dt + \int_1^2 f(t) dt - \int_1^2 1 dt$

$$= \int_0^2 f(t) dt - [t]_1^2 = 5 - [2-1] = 4$$

b)  $mx - y + b = 0$  and using the perpendicular distance formula with  $(x_1, y_1)$  as  $(0, 0)$  and  $a = m, b = -1, c = b, d = 2$  we obtain:

$$2 = \frac{|m(0) - 1(0) + b|}{\sqrt{m^2 + 1}} \Rightarrow 2 = \frac{b}{\sqrt{m^2 + 1}}$$

$$\therefore 4 = \frac{b^2}{m^2 + 1} \Rightarrow m^2 + 1 = \frac{b^2}{4}$$

c) When  $n=1$ ,  $LHS = 1$ ,  $RHS = \frac{1}{6}(1)(3)(2) = 1$

$\therefore$  true for  $n=1$ .

Assume true for  $n=k$

$$\text{i.e. } S_k = \frac{1}{6}k(4k-1)(k+1)$$

Prove true for  $n=k+1$

i.e. Prove that  $S_{k+1} = S_k + T_{k+1}$

$$RHS = \frac{1}{6}k(4k-1)(k+1) + (k+1)(2k+1)$$

$$= \frac{k(4k-1)(k+1) + 6(k+1)(2k+1)}{6}$$

$$= \frac{(k+1)[k(4k-1) + 6(2k+1)]}{6}$$

$$= \frac{(k+1)(4k^2 + 11k + 6)}{6}$$

$$= \frac{1}{6}(k+1)(k+2)(4k+3) = S_{k+1}.$$

Thus proved true for  $n=1$ , assumed true for  $n=k$  and proved true for  $n=k+1$ , so by Mathematical Induction it is true for  $n=1+1=2, n=2+1=3$ , and so on for all positive integers  $n$ .

d) i)  $N = 8000 + Ae^{kt} \Rightarrow Ae^{kt} = N - 8000$

$$\therefore \frac{dN}{dt} = kAe^{kt} = k(N - 8000)$$

ii) When  $t=0, N=25000$

$$\therefore 25000 = 8000 + Ae^0 \Rightarrow A = 17000$$

When  $t=5, N=29250$

$$\therefore 29250 = 8000 + 17000e^{5k}$$

$$e^{5k} = \frac{21250}{17000}$$

$$5k = \ln\left(\frac{21250}{17000}\right) \Rightarrow k = 0.0446$$

iii)  $50000 = 8000 + 17000e^{0.0446t}$

$$e^{0.0446t} = \frac{42000}{17000} \Rightarrow t = 20$$

$\therefore$  In 2020 the population will reach 50 000.

**Question 4.**

a) i) The tangent  $AT$  meets radius  $OA$  at  $90^\circ$ .  
 $\therefore \angle OAT = 90^\circ$ .

Also, a line from the centre that bisects a chord is perpendicular to the chord

$$\therefore \angle OXT = 90^\circ$$

Hence  $AOXT$  is a cyclic quad (opp.  $\angle$ 's supp.)

ii) Since  $AOXT$  is cyclic  $\angle AOT = \angle AXT$   
[Angles at circumference standing on same arc  $AT$ ].

b) i)  $4 - x^2 > 0 \Rightarrow -2 < x < 2$ .

ii)  $A = \int_0^1 \frac{1}{\sqrt{4-x^2}} dx = \left[ \sin^{-1} \frac{x}{2} \right]_0^1$

$$= \sin^{-1}\left(\frac{1}{2}\right) - \sin^{-1}(0) = \frac{\pi}{6} u^2$$

**Question 4( continued ).**

c) i)  $\alpha + \beta + \gamma = -\frac{b}{a} = -\frac{1}{2}$

ii)  $\alpha\beta\gamma = -\frac{d}{a} = 1$

iii)  $\alpha\beta + \alpha\gamma + \beta\gamma = \frac{c}{a} = -\frac{1}{2}$

iv) On Expansion

$$(\alpha-1)(\beta-1)(\gamma-1) = \alpha\beta\gamma - (\alpha\beta + \alpha\gamma + \beta\gamma) + (\alpha + \beta + \gamma) - 1$$

$$= 1 - \left(-\frac{1}{2}\right) + \left(-\frac{1}{2}\right) - 1 = 0$$

**Question 5.**

a)  $\frac{dV}{dt} = 450$ ,  $V = \frac{4}{3}\pi r^3$ ,  $\frac{dV}{dr} = 4\pi r^2$ ,  $r = 15 \text{ cm}$

$\therefore$  using the chain rule  $\frac{dV}{dt} = \frac{dV}{dr} \cdot \frac{dr}{dt}$

$$\therefore 450 = 4\pi r^2 \cdot \frac{dr}{dt} \Rightarrow \frac{dr}{dt} = \frac{450}{4\pi \times 15^2} = \frac{1}{2\pi} \text{ cm/s.}$$

b)  $V = \pi \int_0^{\frac{3\pi}{4}} \sin^2 x \, dx$   $\left[ \cos 2\theta = 1 - 2\sin^2 \theta \right]$

$$\therefore \left[ \sin^2 \theta = \frac{1}{2}(1 - \cos 2\theta) \right]$$

$$\therefore V = \pi \int_0^{\frac{3\pi}{4}} \frac{1}{2}(1 - \cos 2x) \, dx = \frac{\pi}{2} \left[ x - \frac{1}{2} \sin 2x \right]_0^{\frac{3\pi}{4}}$$

$$= \frac{\pi}{2} \left[ \left( \frac{3\pi}{4} - \frac{1}{2} \sin \frac{3\pi}{2} \right) - (0 - 0) \right]$$

$$= \frac{\pi}{2} \left[ \frac{3\pi}{4} + \frac{1}{2} \right] = \frac{3\pi^2 + 2\pi}{8} u^3$$

c) i)  $y = \frac{x^2}{4a} \Rightarrow \frac{dy}{dx} = \frac{x}{2a} \Rightarrow m_T = t$  when  $x = 2at$

$\therefore m_N = -\frac{1}{t} \Rightarrow \text{Eq}^n \text{ of Normal: } y - at^2 = -\frac{1}{t}(x - 2at)$

$\therefore ty - at^3 = 2at - x \Rightarrow x + ty = 2at + at^3$

ii) Using (i) and putting  $x = 0 \Rightarrow y = 2a + at^2$

$\therefore Q$  has coordinates  $(0, 2a + at^2)$

and since  $R$  lies on the parabola then when

$$y = 2a + at^2, \quad x^2 = 4a(2a + at^2) = 4a^2(2 + t^2)$$

which gives  $x = -2a\sqrt{2 + t^2}$  since  $R$  is in the second quadrant.

$\therefore R$  has coordinates  $(-2a\sqrt{2 + t^2}, 2a + at^2)$ .

iii)  $M$  has coordinates  $(-a\sqrt{2 + t^2}, 2a + at^2)$

For the locus of  $M$ :  $x = -a\sqrt{2 + t^2}$

$\therefore x^2 = a^2(2 + t^2) = a(2a + at^2)$

$= ay$

Hence  $x = -\sqrt{ay}$  since  $R$  is in the 2<sup>nd</sup> Quad.

is the locus of  $M$ .

**Question 6.**

a) i)  $\ddot{x} = \frac{d}{dx} \left( \frac{1}{2} v^2 \right)$

$\therefore \frac{d}{dx} \left( \frac{1}{2} v^2 \right) = 9x - 18$

$\therefore \frac{1}{2} v^2 = \frac{9x^2}{2} - 18x + C$

$\frac{1}{2}(-6)^2 = \frac{9(4)^2}{2} - 18(4) + C \Rightarrow C = 18$

$\therefore \frac{1}{2} v^2 = \frac{9x^2}{2} - 18x + 18$

$$v^2 = 9x^2 - 36x + 36$$

$$= 9(x^2 - 4x + 4) = 9(x - 2)^2$$

ii)  $v = -3(x - 2)$  since  $v < 0$  when  $t = 0$

$\frac{dx}{dt} = -3(x - 2) \Rightarrow \frac{dt}{dx} = \frac{-1}{3(x - 2)}$

**Question 6 a) ii) continued.**

$t = -\frac{1}{3} \ln(x - 2) + C$

when  $t = 0, x = 4 \Rightarrow 0 = -\frac{1}{3} \ln 2 + C \Rightarrow C = \frac{1}{3} \ln 2$

$\therefore t = \frac{1}{3} [\ln 2 - \ln(x - 2)] \Rightarrow 3t = \ln \left( \frac{2}{x - 2} \right)$

$\therefore e^{3t} = \frac{2}{x - 2} \Rightarrow x - 2 = \frac{2}{e^{3t}}$

$\therefore x = 2 + 2e^{-3t} \Rightarrow x = 2(1 + e^{-3t})$

b) i) When  $t = 0, x = 0, \dot{x} = V \cos \theta$  (a constant)

$\ddot{x} = 0 \Rightarrow \dot{x} = C \Rightarrow \dot{x} = V \cos \theta$

$\therefore x = Vt \cos \theta + C$  but  $C = 0$  by I.C.

Hence  $x = Vt \cos \theta$ .

ii)  $y = Vt \sin \theta - 5t^2 + 1 \Rightarrow \dot{y} = V \sin \theta - 10t$

At maximum height  $\dot{y} = 0 \Rightarrow t = \frac{V \sin \theta}{10}$

and when  $t = \frac{V \sin \theta}{10}, y = 2$

$\therefore 2 = V \left( \frac{V \sin \theta}{10} \right) \sin \theta - 5 \left( \frac{V \sin \theta}{10} \right)^2 + 1$

$2 = \frac{V^2 \sin^2 \theta}{10} - \frac{5V^2 \sin^2 \theta}{100} + 1$

$200 = 10V^2 \sin^2 \theta - 5V^2 \sin^2 \theta + 100$

$\therefore V^2 \sin^2 \theta = 20 \Rightarrow V^2 = \frac{20}{\sin^2 \theta} \Rightarrow V = \frac{\sqrt{20}}{\sin \theta}$

iii)  $\tan \theta = \frac{9}{40} \Rightarrow \sin \theta = \frac{9}{41}$  using Pythagoras

$\therefore V = \frac{\sqrt{20}}{\frac{9}{41}} = \frac{41\sqrt{20}}{9}$

$\therefore V \approx 20.37 \text{ ms}^{-1}$  correct to 2 decimal places.

**Question 7.**

a)  $\lim_{x \rightarrow 0} \frac{1 - \cos 2x}{3x^2} = \lim_{x \rightarrow 0} \frac{2 \sin^2 x}{3x^2}$

$$= \frac{2}{3} \lim_{x \rightarrow 0} \frac{\sin x}{x} \cdot \frac{\sin x}{x}$$

$$= \frac{2}{3} \cdot 1 \cdot 1 = \frac{2}{3}$$

b) i) Gradient of  $PQ = \tan(180 - \theta) = -\tan \theta$

$\therefore y - 2 = -\tan \theta (x - 1)$

$y - 2 = -x \tan \theta + \tan \theta$

$y = \tan \theta - x \tan \theta + 2$

ii) Using i) when  $x = 0, y = 2 + \tan \theta$

and when  $y = 0, x = \frac{2 + \tan \theta}{\tan \theta}$

$\therefore \text{Area } \Delta OPQ = \frac{1}{2} OP \times OQ$

$$= \frac{1}{2} \left( \frac{2 + \tan \theta}{\tan \theta} \right) (2 + \tan \theta)$$

$$= \frac{1}{2} \left( \frac{4 + 4 \tan \theta + \tan^2 \theta}{\tan \theta} \right)$$

$$= \frac{1}{2} \left( \frac{4}{\tan \theta} + 4 + \frac{\tan^2 \theta}{\tan \theta} \right)$$

$$= \frac{2}{\tan \theta} + 2 + \frac{\tan \theta}{2}$$

iii) In part ii) let  $t = \tan \theta$

$\therefore A = \frac{t}{2} + 2 + \frac{2}{t} = \frac{t}{2} + 2 + 2t^{-1}$

For a minimum area  $\frac{dA}{dt} = 0$  and  $\frac{d^2A}{dt^2} > 0$

$\frac{dA}{dt} = \frac{1}{2} - 2t^{-2} = \frac{1}{2} - \frac{2}{t^2}$  and  $\frac{d^2A}{dt^2} = \frac{4}{t^3}$

$\therefore \frac{2}{t^2} = \frac{1}{2} \Rightarrow t^2 = 4 \Rightarrow t = 2$ , also  $\frac{d^2A}{dt^2} = \frac{1}{2}$

$\therefore$  minimum area when  $\tan \theta = 2$

iv)  $\therefore \text{Minimum Area} = 1 + 2 + 1 = 4 u^2$ .